

CORE 200 Scientific Methods

Handout 10

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Reflect and discuss (A)

 1.

 2.

Read the passage below, making note of any important points and definitions.

CRITICAL DISCOURSE

...[We] saw various ways and techniques how science defends its knowledge claims in a much more systematic way than we do in our life. However, these ways and techniques can only function properly and effectively if they are realized by people who are socialized in a certain way and further embedded in fitting social organizations. Roughly speaking, the social norms and the social institutions that constitute the social organization of scientific communities must be conducive to the exertion of the cognitive norms that must be operative for the enterprise to

reach its institutional goal.

The aspect of the social organization of science I will focus on here is institutionalized critical discourse. The term “critical” signifies the goal of probing claims, the term “discourse” signifies the involvement of various members of the community, and the term “institutionalized” signifies some sort of social organization and order, hence some sort of systematicity.

NORMS AND INSTITUTIONS

The norms relevant for the specific functioning of scientific communities have been discussed in the sociology of sci-

ence starting with Robert Merton's classic paper "Science and the Social Order," first published in 1938, and especially in Merton's "The Normative Structure of Science," first published in 1942. Merton identified four norms, also called "institutional imperatives," that are binding for scientists because they implement the institutional goal of science. The institutional goal of science is, in Merton's words, "the extension of certified knowledge." The four norms are universalism, "communism," disinterestedness, and organized skepticism. The first three are less interesting in our context so I will not discuss them further. The fourth factor, however, is relevant to our discussion. Merton's "organized skepticism" is usually seen as the imperative to emphasize "primarily an institutionally enjoined critical attitude toward the work of fellow scientists."

[P]hilosopher Helen Longino has fleshed out what "critical discursive interactions" are. They "are social processes of knowledge production. They determine what gets to remain in the public pool of information that counts as knowledge." In particular, Longino identifies four features that are "necessary to assure the effectiveness of discursive interactions," i.e., critical discourse:

- i) Venues: "publicly recognized forums for the criticism of evidence, of methods, and of assumptions and reasoning";
- ii) Uptake: "community members pay attention to and participate in the critical discussion taking place";
- iii) Public standards: "There must be publicly recognized standards by reference to which theories, hypotheses, and observational practices are evaluated";
- iv) Tempered equality: "equality" here means that criticism of all members of the community is admitted, and "tempered" means that there is some sort of weighing according to intellectual authority. What that precisely means is difficult but without further relevance for us.

PRACTICES IN SCIENCE FOSTERING CRITICAL DISCOURSE

Let us start with publications. Publications are an obvious precondition for an open scientific discourse in which every member of the community can participate. Thus, there is a sort of imperative for scientists to publish their results. Typically, the journals and presses that publish academic work have a reviewing system in which the quality of the submitted work is assessed-the so-called peer review system. This system has two functions. First, it acts as a critical filter by eliminating work that is assessed as unsuitable by the reviewers and the editors because of either quality issues or an unfitting subject matter. In many cases, authors receive a review report in which the refusal

is argued. In the most prestigious journals, the rejection rate is rather high, typically more than 90 percent. Second, the peer review system helps to increase the quality of scientific work in cases of "conditional acceptance" or "suggested resubmission." In these cases, changes in the original manuscript are suggested and argued in order to improve it. The reviewing process is mostly "blind" in the sense that the author does not know who the reviewers are. Often, it is even "double-blind" in the sense that also the reviewers do not know who the author is. Clearly, the latter practice is designed to avoid positive or negative bias due to known authorship. Academic presses use similar procedures as the journals for the publication of books.

For discussion of the work of others, there are many established channels. Conference talks or invited talks are invariably followed by a discussion period, often introduced by a commentator who studied the paper beforehand. For special topics, panel discussions are organized at conferences. Journals often have a particular section of shorter discussion papers that take up issues typically published earlier in the same journal. Book reviews that not only inform potential readers about the work but also critically assess it are a regular part of many journals; there are even journals that exclusively publish book reviews. In addition, the current state of play in some fields is critically summarized and assessed in review articles that are published in special sections of regular scientific journals or in specific review journals.

This sketchy overview makes clear that there are many platforms established in science on which critical discourse may and indeed does take place. One additional feature is remarkable. As already emphasized by Robert Merton, there are no taboos regarding subjects to be discussed in science. Of course, there may be quite touchy subjects, for whatever reason, be it political, ideological, scientific, or otherwise, and people without a permanent position at a university are well advised to exert great caution when approaching such subjects. However, in science, one will hardly find the rejection of some contribution for the officially declared reason that such a subject must not be discussed. Officially, there are no such taboos.

In the natural sciences, the institutions of critical discourse just discussed have been in place, at least in rudimentary form, since the seventeenth century. From the nineteenth century on, also the humanities and the then emerging social sciences have developed similar mechanisms. From the mid-twentieth century on, novel institutions have been developed for a novel brand of research, so-called big science.

Excerpted from Paul Hoyningen-Huene. *Systematicity. The Nature of Science*. Oxford University Press, 2013



Reflect and discuss (B)



3.



4.



5.



6.

Read the passage below, making note of any important points and definitions.

SOCIAL SCIENCES

Philosophical interest in the foundations of the social sciences has a long history, dating back at least to Mill's influential work on the social sciences. Some foundational issues have also been systematically discussed by social scientists themselves, such as Durkheim (1895/1966) and Weber (1903/1949). Around the middle of the twentieth century, the social sciences again received serious philosophical attention. The focus was largely on their being human sciences and the philosophical issues this raised. More recently, philosophers have directed their attention to the different social sciences in their own right, especially economics.

A central focus of discussion is whether the social sciences are fundamentally different from the natural sciences. Logical empiricists attempted to incorporate the social sciences into their models for the natural sciences (see Unity of Science Movement). Others have argued that the social sciences are unique. This has framed many of the debates within the philosophy of the social sciences, a number of which are briefly discussed in what follows:

- i) Are There Social Science Laws? Laws played important roles in empiricist accounts of explanation, theories, confirmation, and prediction, but it is unclear whether there are laws of the social sciences (see Laws of Nature). Social phenomena are complex, involve reference to social kinds, and require idealizations. As a result, many argue that generalizations of the social sciences, if they are laws at all, require ineliminable *ceteris paribus* clauses. Others argue that the social sciences ought not even attempt to create generalizations or grand theories, as social phenomena are essentially historical and local.
- ii) Do Social Scientific Theories Yield Testable Predictions? Because of the complexity of social systems, social scientific theories require idealizations. Given the nature of these idealizations, deriving empirical predictions from social scientific theories is difficult at best (see Prediction). As a result, many argue that social scientific theories are not testable. This is exacerbated by the reflexive nature of social science theories: the very act of theorizing can change the behavior one is theorizing about. Moreover, if human action is explained by agents' desires and beliefs, social scientists seem to be caught in an intentional circle, making it difficult to derive any testable claims (see Rosenberg 1988).
- iii) Is the Methodology of the Social Sciences Distinct? Given that social sciences involve humans and human behavior on a large scale, experimentation has not played a significant role in the social sciences (see Experimentation). There are also many who question whether the social sciences can be naturalized.

Some argue that understanding social action is essentially a hermeneutic enterprise, distinctly different from the natural sciences.

- iv) What Are the Ontological Commitments of Scientific Theories? Beginning with Mill and, subsequently, Durkheim and Weber, there have been debates as to whether social scientific theories are reducible to theories about individual behavior (see Methodological Individualism). Moreover, after Nagel's influential account of intertheoretic reduction, it has been argued that social phenomena are multiply realizable, and therefore, social science theories are not reducible to lower-level theories (see Emergence; Reductionism; Supervenience). Additionally, given that social scientific theories involve idealizations, there are questions about whether these theories ought to be interpreted realistically or instrumentally (see Instrumentalism; Realism).
- v) What Is the Nature of Social Scientific Explanations? Some, such as Hempel (1962), have argued that social scientific explanations are no different than in the physical sciences. Others, however, have questioned this. If there are no social scientific laws, then social scientific explanation cannot be captured by the covering law model (see Explanation). Social sciences also often rely on functional explanations, which, while similar to biology, seem to be different from explanations in physics (see Function). Others, following Winch (1958), have argued that social sciences explain action, not behavior, which requires understanding the meaning of the action (not its causes), and therefore must include the actors' intentions and social norms. Moreover, some have argued that actions are governed by reasons, and are therefore not susceptible to causal explanation, a view that was later convincingly refuted by Davidson (1963). An alternative account of how beliefs and desires can explain actions has been formalized in rational choice theory (see Decision Theory), although there are questions about whether such explanations capture how people actually behave, rather than how they ought to behave.
- vi) What Is the Relationship Between Social Science and Social Values? There has also been concern with the connection between social values and the social sciences. Taylor (1971), for example, argues that social theory is inherently value-laden, and Habermas (1971) argues that social theory ought to engage in social criticism.

Sahotra Sarkar and Jessica Pfeifer, eds. *The philosophy of science: an introduction*. Routledge, 2006.